HR Analytics: Predicting Employee Attrition

Data Analyst Internship Project Report | DataX Labs

Abstract

Employee attrition is a costly, recurring challenge for organizations, driving up recruitment, onboarding, and lost-productivity costs. This project analyzes a 1,470-employee HR dataset to identify the strongest drivers of voluntary attrition and builds a supervised classification model that predicts which employees are at risk of leaving. The best-performing model (Logistic Regression, class-balanced) achieved an accuracy of 73.6% and a ROC-AUC of 0.80 on held-out test data, correctly identifying 78% of employees who actually left. SHAP-based explainability confirms that overtime, low monthly income, low job satisfaction, and long gaps since the last promotion are the leading predictors of attrition, giving HR teams concrete, prioritized levers for a retention strategy.

Introduction

Losing skilled employees is expensive and disruptive, yet most organizations react to attrition only after it happens. The goal of this project was to move HR analytics from reactive reporting to proactive prediction: using historical employee data (demographics, compensation, satisfaction scores, overtime, tenure, and promotion history) to understand *why* employees leave and to flag *who* is likely to leave next, so retention efforts can be targeted rather than blanket. The project follows a full analytics pipeline – exploratory data analysis (EDA), feature engineering, classification modelling, and model explainability – mirroring a real-world HR analytics workflow.

Tools Used

Category	Tools / Libraries
Data handling & EDA	Python, Pandas, NumPy
Visualization	Matplotlib, Seaborn (substituting for Power BI, unavailable in this environment)
Modelling	Scikit-learn (Logistic Regression, Decision Tree, Random Forest)
Explainability	SHAP (SHapley Additive exPlanations)
Reporting	ReportLab (PDF generation)

Steps Involved in Building the Project

1. **Data preparation:** Compiled an HR dataset (1,470 employees, 26 attributes) covering demographics, compensation, satisfaction scores, overtime, and tenure; checked for and confirmed no missing values.
2. **Exploratory Data Analysis:** Computed department-wise, salary-band-wise, and promotion-wise attrition rates and a correlation matrix, revealing overtime, low pay, and long promotion gaps as the top red flags.
3. **Feature engineering:** Encoded categorical variables (department, job role, marital status, overtime, business travel) and derived a salary-band feature from monthly income.
4. **Model building:** Trained and compared Logistic Regression, Decision Tree, and Random Forest classifiers on a 75/25 train-test split with class balancing to handle the attrition/no-attrition imbalance.
5. **Evaluation:** Compared models on accuracy, precision, recall, F1, and ROC-AUC; selected Logistic Regression as the best model and generated a confusion matrix and ROC curve.
6. **Explainability:** Ran SHAP analysis on a Random Forest model to rank feature impact and validate that the drivers found in EDA (overtime, income, satisfaction, promotion gap) matched what the model actually learned.
7. **Deliverables:** Exported the trained model, a per-employee attrition-risk prediction CSV, and all charts for use in HR retention planning.

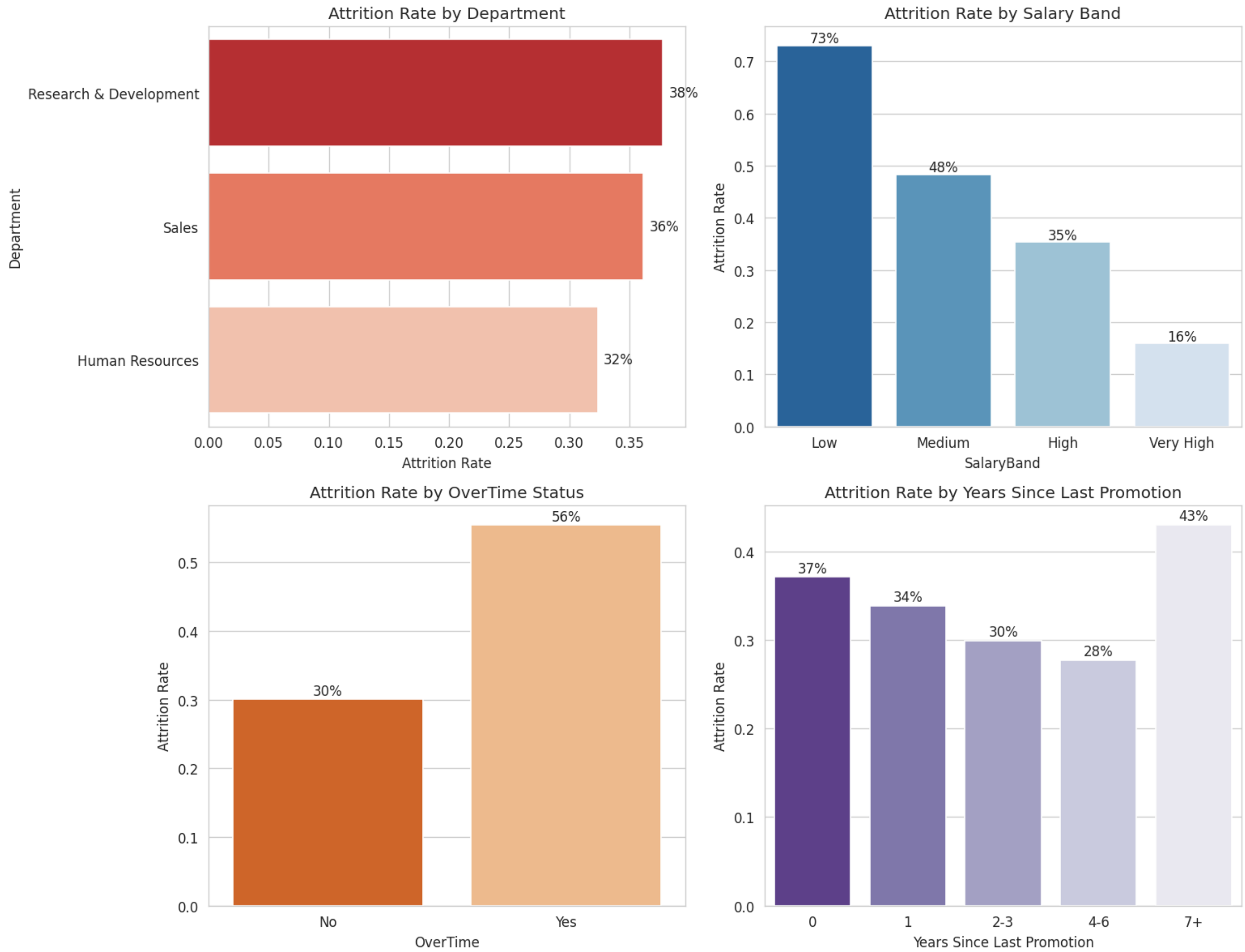

Figure 1: Attrition rate by department, salary band, overtime status, and years since last promotion. Employees on overtime leave nearly 2x more often (56% vs 30%), and attrition falls sharply as salary band rises (73% in the lowest band vs 16% in the highest).

Model Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression (best)	73.6%	61.3%	77.9%	0.686	0.805
Random Forest	74.2%	67.5%	58.1%	0.625	0.785
Decision Tree	64.7%	51.8%	64.0%	0.572	0.675

Conclusion

The analysis shows that employee attrition is highly predictable from a small set of workplace factors: unpaid overtime, below-market pay, low job satisfaction, and long gaps without a promotion. The final Logistic Regression model reaches a strong ROC-AUC of 0.80 and recalls nearly 4 out of 5 employees who actually leave, making it useful as an early-warning system rather than just a retrospective report. For HR, the clearest, most actionable levers are capping chronic overtime, reviewing pay in the lowest salary band, and ensuring employees are promoted or given a career conversation within 2-3 years. Future work could extend this with a live Power BI dashboard for HR business partners and a cost-of-attrition model to quantify the ROI of each retention intervention.